Gredu: Transforming schools through digital processes and data-driven insights on Google Cloud

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Gredu migrates t
Run to improve d
productivity and s
data pipelines on
empowering edu
stakeholders witl
driven insights th
student learning.

About Gredu

Pronounced as "Gredu," taken from a word 'Credo' which means "believe" in Welsh language, it was founded in 2017 to improve student performance by digitally transforming the learning environments of schools on its educational technology platform. As a solution for teachers, parents, students, and administrators, Gredu provides features such as curriculum-based learning content, interactive classes, and e-libraries to reduce administration and increase school transparency. Through seminars, Gredu raises awareness about bullying

Google Cloud results

- Facilitates real-time collaboration between schools, teachers, parents, and students with seamless data transfer using Cloud Pub/Sub
- Automates marking so teachers can provide timely feedback to parents and students
- Builds machine learning algorithms with BigQuery to correlate students' online learning

Scale: down traffic during

among students, creating a safe learning environment.

Industries: Education

Location: Indonesia

Pub/Sub (https://cloud.google.com/pubsub/),

Google Cloud (https://cloud.google.com/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud Run (https://cloud.google.com/run/)

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Cloud Pub/Sub (https://cloud.google.com/pubsub/)

Firebase (https://firebase.google.com/)

behaviors
and
academic
performance

Although a teacher's core responsibility is to teach students, the reality is the majority of their working time is consumed by activities other than teaching. According to OECD, teachers, on average, spend 44% of their time on teaching (https://www.oecd-ilibrary.org/docserver/d4187f9b-en.pdf?expires=1649762198&id=id&accname=guest&checksum=244882A691807BD1803A3ADB4D8B7D80). At the same time, non-teaching tasks, such as lesson planning, marking homework, and parent communication, take up a large proportion of a teacher's working time. Gredu (https://gredu.asia/), an ed-tech platform, simplifies administrative tasks, so teachers can do what they do best, teach.

Gredu, which runs on Google Cloud (https://cloud.google.com/), facilitates collaboration between schools, teachers, parents, and students on its school management platform. Schools can integrate Gredu and improve the learning process through several subscription options, including a premium package with a custom-branded platform.

"Our vision is to transform

schools with digital processes on Google Cloud. Through our platform, teachers and schools can become more productive and data-driven so they can help students improve."

—Arya Budi Nugraha, Co-Founder and Chief Commercial Partnership, Gredu

With Gredu Teacher (https://play.google.com/store/apps/details?id=co.gredu.teacher&hl=en_IN&gl=US), teachers can organize lesson plans, track student performance, and communicate with parents. Gredu Parent (https://play.google.com/store/apps/details?id=co.gredu.parent&hl=en_IN&gl=US) helps parents monitor their child's learning progress and receive invoices regarding their child's schooling. For students, the Gredu Student app (https://play.google.com/store/apps/details?id=co.gredu.student&hl=en_IN&gl=US) is a virtual classroom where they can submit assignments and attend online classes. The app enabled more than 350,000 students to continue their education at home when schools were closed during the COVID-19 pandemic.

"Our vision is to transform schools with digital processes on Google Cloud," says Arya Budi Nugraha, Co-Founder and Chief Commercial Partnership at Gredu. "Through our platform, teachers and schools can become more productive and data-driven so they can help students improve."

Optimizing time and costs with Cloud Run to speed up development

Initially, Gredu built its app by choosing the native apps and utilizing Firebase (https://firebase.google.com/). Mainly for open-source infrastructure and the mobile apps development. As

Gredu’s user base and application features grew, its lean engineering team found it increasingly difficult to keep up with development demands and manage infrastructure. DevOps, for example, manually added more containers or VMs to handle spikes in usage, particularly during exams. Gredu purchased standby instances to cope with 5X higher than average loads to avoid performance issues.

After spending sufficient time exploring further on serverless method, Gredu migrated main portion of its application to Cloud Run (https://cloud.google.com/run), a fully managed, serverless platform, to increase resources on demand without over-provisioning servers. Gredu also uses BigQuery (https://cloud.google.com/bigquery) to analyze education data better.

"BigQuery, Cloud Run, and Cloud Pub/Sub (https://cloud.google.com/pubsub/docs/overview) provide the critical backbone for a seamless flow of data between teachers, students, parents, and schools, so they can communicate in real time," says Mohammad Fachri, Co-Founder and Chief Technology Officer at Gredu. "As a business, we optimize costs by only paying for cloud resources actually used, not idle capacity. Plus, developers have more time to focus on building new features such as Gredu Pocketbook and Gredu Assistant, instead of managing servers."

According to Fachri, a simple interface and seamless integration between Google Cloud products make it ideal for its developers.

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Analyzing student performance for better results

Report cards used to be the only means for evaluating a student's academic performance. Occasionally, a teacher can make a mistake in marking exams and calculating grades on the report card. With Gredu, exams are marked automatically, reducing manual errors.

With time saved from marking papers, teachers can pay more attention to their students and tailor teaching based on their strengths and weaknesses. Gredu uses BigQuery to analyze data across the platform so teachers can gain a holistic view of each student's learning journey in real-time. For example, if a student's poor performance coincides with absences, a teacher may need to give additional coaching and collaborate with parents to improve attendance.

Gredu recognizes that students want learning to be fun and thus, introduced the Pocketbook feature in the student app. As students move from one learning level to another, this feature allows them to test their mastery through question-based tasks before advancing to the next. User interaction with this feature generates data that provides Gredu with insight into students' learning patterns.

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—Mohammad Fachri, Co-Founder and Chief Technology Officer, Gredu

Driving personalized learning with AI to improve student success

Students learn in different ways. For example, visual learners learn by seeing charts and picture aids, while auditory learners prefer discussions and verbal instructions. Even though teachers have years of experience teaching, they cannot analyze vast amounts of education data as Gredu can to understand how students learn. For teachers to gain such insights, Gredu will expand its new data pipelines in 2022, including creating a metacognitive learning algorithm.

By integrating structured data, such as results for completed assignments, with unstructured data, such as online discussions with teachers and classmates, this machine learning algorithm analyzes students' question-solving behavior. Gredu helps teachers by recommending content in real-time that adapts to each student’s learning style.

"Unstructured data was a key driver for adopting BigQuery. With a data lake on BigQuery, we don’t need to worry about the schema or structure of data," says Fachri. "BigQuery brings the ability to extract, transform, and load data in a scalable and efficient manner to power real-time education insights as more schools join the platform."